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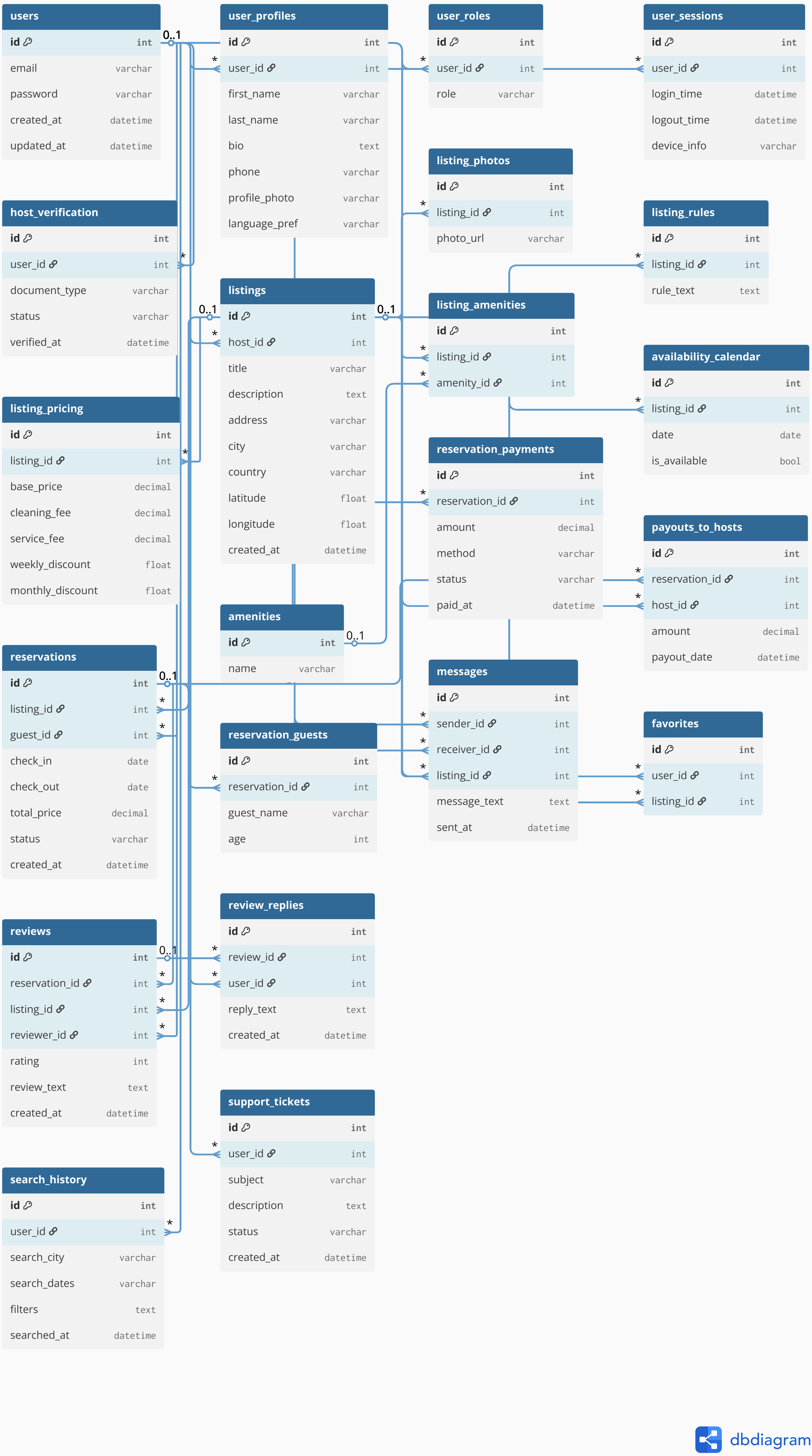
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user_profiles
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last_name varchar
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phone varchar
profile_photo varchar
language_pref varchar
user_roles
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role varchar
user_sessions
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status varchar
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listings
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city varchar
country varchar
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longitude float
created_at datetime
listing_amenities
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listing_id int
amenity_id int
availability_calendar
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listing_id int
date date
is_available bool
listing_pricing
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listing_id int
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cleaning_fee decimal
service_fee decimal
weekly_discount float
monthly_discount float
reservation_payments
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reservation_id int
amount decimal
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status varchar
paid_at datetime
payouts_to_hosts
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reservation_id int
host_id int
amount decimal
payout_date datetime
amenities
id int
name varchar
reservations
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listing_id int
guest_id int
check_in date
check_out date
total_price decimal
status varchar
created_at datetime
messages
id int
sender_id int
receiver_id int
listing_id int
message_text text
sent_at datetime
favorites
id int
user_id int
listing_id int
reservation_guests
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guest_name varchar
age int
review_replies
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review_id int
user_id int
reply_text text
created_at datetime
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rating int
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created_at datetime
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created_at datetime
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user_id int
search_city varchar
search_dates varchar
filters text
searched_at datetime
0..1
*
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AIRBNB-Replica Requirements Specification

Introduction

This platform is an Airbnb-style accommodation system that enables users to search, book, and list properties. The system captures real-world interactions between guests and hosts, translating them into structured data, tables, and relational constraints.

The system revolves around two core user journeys: guests finding and booking places to stay, and hosts listing and managing their properties. All interactions — searching, listing, booking, reviewing, and communicating — generate relational data that maintains platform integrity.

Roles (User Groups)

The system defines four distinct roles, each with specific responsibilities and data access requirements:

1. Guest — A person searching for accommodations. Guests browse listings, make bookings, process payments, leave reviews, and communicate with hosts.

2. Host — A property owner or manager who creates listings and manages availability. Hosts respond to booking requests, receive payments, reply to guest inquiries, and moderate their property information.

3. Admin — A platform administrator who maintains system integrity. Admins oversee listings, handle dispute resolution, manage reports, and ensure data quality across the platform.

4. External Services — Third-party providers such as payment processors and identity verification services that influence system data through integrations.

Actions

Guest Actions

- Create account and authenticate (users table)
- Search for listings with filters: location, dates, price, amenities (search_history table)
- Book accommodations and specify dates (reservations table)
- Process payments for reservations (reservation_payments table)
- Leave reviews and ratings for completed stays (reviews table)
- Send and receive messages with hosts (messages table)
- Save favourite listings for later reference (favorites table)

Host Actions

- Create and publish property listings with descriptions, photos, and pricing (listings table)
- Upload and manage property photos (listing_photos table)
- Set and update availability calendars (availability_calendar table)
- Accept or decline booking requests (updates reservations table)
- Receive payouts for completed bookings (payouts_to_hosts table)
- Respond to guest messages and inquiries (messages table)
- Manage house rules per property (listing_rules table)

Admin Actions

- Review and approve or reject new listings for policy compliance
- Investigate and resolve disputes between guests and hosts (support_tickets table)
- Handle user reports regarding inappropriate content or behaviour
- Monitor platform data integrity and identify anomalies
- Enforce community standards and terms of service

Data and Functions

Core Data Tables

users — Stores guest, host, and admin accounts with authentication credentials. Central entity referenced by nearly all other tables.

user_profiles — Extends the users table with personal details such as name, bio, phone number, and profile photo.

user_roles — Tracks whether a user is acting as a guest, host, or admin; supports multi-role accounts.

user_sessions — Logs login and logout events per device for auditing and security purposes.

host_verification — Records identity document submissions and their approval status for host accounts.

listings — Stores property details including title, description, location coordinates, and host reference.

listing_photos — Stores multiple images per listing, each with a URL path.

listing_rules — Captures house rules defined by the host per property.

listing_pricing — Records base price, fees, and discount tiers (weekly and monthly) for each listing.

amenities — A reference table of all available amenity types supported by the platform.

listing_amenities — Ternary junction table associating listings with their amenities; enables many-to-many mapping between listings and amenity types via the host.

availability_calendar — Tracks which specific dates are available or blocked per listing.

reservations — Records full booking details: guest, listing, dates, total price, and status workflow.

reservation_guests — Ternary table capturing additional travellers on a reservation, linked to both the reservation and the listing.

reservation_payments — Ternary table logging each payment transaction, linking the reservation, guest, and payment method with amount and status.

payouts_to_hosts — Tracks disbursements to hosts after guest check-in, referencing the reservation and host.

reviews — Stores post-stay ratings and written feedback linked to a specific reservation, listing, and reviewer.

review_replies — Allows hosts to post a written response to a guest review.

messages — Records all direct communications between users, threaded by listing context.

favorites — Links guests to listings they have saved for future reference.

search_history — Captures search queries and applied filters for analytics and personalised recommendations.

support_tickets — Logs platform support requests submitted by any user, with status tracking.

Ternary Relationships

The model includes three explicit ternary relationships, where three entities participate simultaneously in a single association:

1. listing_amenities (Listing × Amenity × Host) — A host configures specific amenities for a specific listing. This three-way relationship ensures amenity assignments are scoped to both the property and its owner, preventing cross-host data contamination.

2. reservation_guests (Reservation × Guest × Listing) — Additional travellers are associated with both the booking reservation and the property being stayed in, enabling accurate occupancy tracking per listing.

3. reservation_payments (Reservation × Guest × Payment Method) — Each payment transaction ties the financial event to the guest, the specific reservation, and the payment instrument used, satisfying audit and refund traceability requirements.

Core Functions

Search & Discovery — Full-text search with filters (location, dates, price range, amenities), result ranking by relevance and availability.

Booking Management — Availability checking, reservation creation, date conflict prevention, and booking status workflow (requested → confirmed → completed).

Payment Processing — Secure payment collection, escrow holding, automated host payouts after guest check-in, and refund handling for cancellations.

Review System — Post-stay review submission with ratings (1–5 stars) and mutual review capability between guests and hosts.

Messaging — Direct communication between guests and hosts, threaded by listing context.

Data Integrity — Referential constraints ensuring all bookings reference valid listings and users, with cascade rules and transaction management for payment operations.

Summary

This requirements specification defines a four-role system (Guest, Host, Admin, External Services) with clear action boundaries. The data model supports full Airbnb platform functionality through 22 interrelated tables covering the complete booking lifecycle — from property discovery and availability management through payment processing, review submission, and dispute resolution. Three explicit ternary relationships (listing_amenities, reservation_guests, reservation_payments) capture multi-entity interactions that binary relationships cannot represent. The architecture enforces data consistency through referential constraints while enabling the two-sided marketplace dynamics central to the platform.

Data Dictionary

The following table documents all attributes across all 22 entities, including data types and descriptions. Primary keys (PK) and foreign keys (FK) are indicated in the data type column.

Entity	Attribute	Data Type	Description
users	id	INT (PK)	Unique identifier for each registered user
users	email	VARCHAR(255)	User login email address, must be unique
users	password	VARCHAR(255)	Hashed password for authentication
users	created_at	DATETIME	Timestamp when the account was created
users	updated_at	DATETIME	Timestamp of the last account update
user_profiles	id	INT (PK)	Unique identifier for the profile record
user_profiles	user_id	INT (FK)	References users.id; links profile to account
user_profiles	first_name	VARCHAR(100)	User's first name
user_profiles	last_name	VARCHAR(100)	User's last name
user_profiles	bio	TEXT	Short free-text self-description
user_profiles	phone	VARCHAR(30)	Contact phone number (hosts required)
user_profiles	profile_photo	VARCHAR(500)	URL path to uploaded profile image
user_profiles	language_pref	VARCHAR(10)	Preferred UI language code (e.g. en, de)
user_roles	id	INT (PK)	Unique identifier for the role record
user_roles	user_id	INT (FK)	References users.id
user_roles	role	VARCHAR(20)	Role type: guest, host, or admin
user_sessions	id	INT (PK)	Unique session identifier
user_sessions	user_id	INT (FK)	References users.id
user_sessions	login_time	DATETIME	Timestamp when session started
user_sessions	logout_time	DATETIME	Timestamp when session ended (nullable)
user_sessions	device_info	VARCHAR(255)	Browser/device metadata for the session
host_verification	id	INT (PK)	Unique verification record identifier
host_verification	user_id	INT (FK)	References users.id of the host
host_verification	document_type	VARCHAR(50)	Type of ID document submitted (e.g. passport)

Entity	Attribute	Data Type	Description
host_verification	status	VARCHAR(20)	Verification status: pending, approved, rejected
host_verification	verified_at	DATETIME	Timestamp of verification approval (nullable)
listings	id	INT (PK)	Unique listing identifier
listings	host_id	INT (FK)	References users.id of the property host
listings	title	VARCHAR(255)	Short descriptive title of the property
listings	description	TEXT	Full property description shown to guests
listings	address	VARCHAR(255)	Street address of the property
listings	city	VARCHAR(100)	City in which the property is located
listings	country	VARCHAR(100)	Country in which the property is located
listings	latitude	FLOAT	GPS latitude coordinate for map display
listings	longitude	FLOAT	GPS longitude coordinate for map display
listings	created_at	DATETIME	Timestamp when the listing was published
listing_photos	id	INT (PK)	Unique photo record identifier
listing_photos	listing_id	INT (FK)	References listings.id
listing_photos	photo_url	VARCHAR(500)	URL path to the uploaded property image
listing_rules	id	INT (PK)	Unique rule record identifier
listing_rules	listing_id	INT (FK)	References listings.id
listing_rules	rule_text	TEXT	Free-text description of a house rule
listing_pricing	id	INT (PK)	Unique pricing record identifier
listing_pricing	listing_id	INT (FK)	References listings.id
listing_pricing	base_price	DECIMAL(10,2)	Nightly base price in platform currency
listing_pricing	cleaning_fee	DECIMAL(10,2)	One-time cleaning fee charged per booking
listing_pricing	service_fee	DECIMAL(10,2)	Platform service fee applied per booking
listing_pricing	weekly_discount	FLOAT	Percentage discount for week-long stays
listing_pricing	monthly_discount	FLOAT	Percentage discount for month-long stays
amenities	id	INT (PK)	Unique amenity type identifier

Entity	Attribute	Data Type	Description
amenities	name	VARCHAR(100)	Name of the amenity (e.g. WiFi, Pool)
listing_amenities	id	INT (PK)	Unique record for the listing-amenity association
listing_amenities	listing_id	INT (FK)	References listings.id
listing_amenities	amenity_id	INT (FK)	References amenities.id
availability_calendar	id	INT (PK)	Unique availability record identifier
availability_calendar	listing_id	INT (FK)	References listings.id
availability_calendar	date	DATE	Specific calendar date for the listing
availability_calendar	is_available	BOOLEAN	True if date is open for booking
reservations	id	INT (PK)	Unique reservation identifier
reservations	listing_id	INT (FK)	References listings.id
reservations	guest_id	INT (FK)	References users.id of the booking guest
reservations	check_in	DATE	Guest check-in date
reservations	check_out	DATE	Guest check-out date
reservations	total_price	DECIMAL(10,2)	Total cost charged for the reservation
reservations	status	VARCHAR(20)	Booking status: pending, confirmed, cancelled, completed
reservations	created_at	DATETIME	Timestamp when the reservation was made
reservation_guests	id	INT (PK)	Unique record for additional guest on a reservation
reservation_guests	reservation_id	INT (FK)	References reservations.id
reservation_guests	guest_name	VARCHAR(150)	Full name of the additional guest
reservation_guests	age	INT	Age of the additional guest
reservation_payments	id	INT (PK)	Unique payment transaction identifier
reservation_payments	reservation_id	INT (FK)	References reservations.id
reservation_payments	amount	DECIMAL(10,2)	Amount charged in this payment transaction
reservation_payments	method	VARCHAR(50)	Payment method used (e.g. credit_card, PayPal)
reservation_payments	status	VARCHAR(20)	Payment status: pending, completed, refunded
reservation_payments	paid_at	DATETIME	Timestamp of successful payment (nullable)
payouts_to_hosts	id	INT (PK)	Unique payout record identifier
payouts_to_hosts	reservation_id	INT (FK)	References reservations.id

Entity	Attribute	Data Type	Description
payouts_to_hosts	host_id	INT (FK)	References users.id of the receiving host
payouts_to_hosts	amount	DECIMAL(10,2)	Net amount disbursed to the host
payouts_to_hosts	payout_date	DATETIME	Timestamp when payout was transferred
reviews	id	INT (PK)	Unique review identifier
reviews	reservation_id	INT (FK)	References reservations.id
reviews	listing_id	INT (FK)	References listings.id
reviews	reviewer_id	INT (FK)	References users.id of the review author
reviews	rating	INT	Numeric rating given by reviewer (1–5)
reviews	review_text	TEXT	Written feedback provided by the reviewer
reviews	created_at	DATETIME	Timestamp when the review was submitted
review_replies	id	INT (PK)	Unique reply identifier
review_replies	review_id	INT (FK)	References reviews.id
review_replies	user_id	INT (FK)	References users.id of the replying user
review_replies	reply_text	TEXT	Host's written response to the review
review_replies	created_at	DATETIME	Timestamp when the reply was submitted
messages	id	INT (PK)	Unique message identifier
messages	sender_id	INT (FK)	References users.id of the message sender
messages	receiver_id	INT (FK)	References users.id of the message recipient
messages	listing_id	INT (FK)	References listings.id; links message to a property
messages	message_text	TEXT	Body content of the message
messages	sent_at	DATETIME	Timestamp when the message was sent
favorites	id	INT (PK)	Unique favourite record identifier
favorites	user_id	INT (FK)	References users.id of the guest
favorites	listing_id	INT (FK)	References listings.id of the saved property
search_history	id	INT (PK)	Unique search history record identifier
search_history	user_id	INT (FK)	References users.id
search_history	search_city	VARCHAR(100)	City entered in the search query

Entity	Attribute	Data Type	Description
search_history	search_dates	VARCHAR(50)	Date range entered in the search query
search_history	filters	TEXT	JSON-serialised filter parameters applied
search_history	searched_at	DATETIME	Timestamp when the search was performed
support_tickets	id	INT (PK)	Unique support ticket identifier
support_tickets	user_id	INT (FK)	References users.id of the ticket submitter
support_tickets	subject	VARCHAR(255)	Short subject line of the support request
support_tickets	description	TEXT	Detailed description of the issue reported
support_tickets	status	VARCHAR(20)	Ticket status: open, in_progress, resolved, closed
support_tickets	created_at	DATETIME	Timestamp when the ticket was submitted